KILLARI DHANUNJAY

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M.Tech in Power Electronics and Drives
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EDUCATION

•National Institute of Technology, Warangal *2025–Present*
M.Tech in Power Electronics and Drives CGPA: 8.18

•Sagi Ramakrishnam Raju Engineering College, Bhimavaram *2020–2024*
B.Tech in Electrical and Electronics Engineering CGPA: 8.11

•Sri Aditya Junior College, Narsapuram *2018–2020*
Board of Intermediate Education, Andhra Pradesh CGPA: 9.7

•Sri Gowthami School, Narsapuram *2018*
Board of Secondary Education, Andhra Pradesh CGPA: 9.8

AREAS OF INTEREST

- Power Electronics and Converter Design, Electric Drives and Motor Control, Digital Control Systems, Power Semiconductor Devices, Renewable Energy Systems, Microgrids

MAJOR PROJECTS

- **M.Tech Dissertation Title: Modeling and Control of a PMSM Drive** *(2026 – Present)*
  Guide: *Dr. Sharmili Das, NIT Warangal*
  Keywords: PMSM, MATLAB/Simulink, Hysteresis Current Control, MPC, PI Control, FOC
  Completed Work:
  - Developed and verified the PMSM drive model and control logic in MATLAB/Simulink.
  - Implemented and evaluated hysteresis current control, Model Predictive Control (MPC), and PI control through simulation.
  - Analyzed current tracking and dynamic response of the implemented control strategies.

  Ongoing and Future Work:
  - Developing a hardware setup using the TI F28379D DSP for real-time PMSM drive operation.
  - Planning hardware implementation of Field-Oriented Control (FOC), including real-time current control and PWM generation.

COURSE PROJECTS

- **Space Vector PWM (SVPWM) Implementation for Three-Phase Inverter** *Power Electronics*
  - Developed and simulated SVPWM switching equations and PWM techniques for a three-phase, two-level voltage source inverter using MATLAB/Simulink.
  - Configured ePWM modules for three-phase switching pulse generation with complementary outputs and dead-time insertion.
- **DC Drive Modeling** *Electric Drives*
  - Developed and simulated a DC drive model using MATLAB/Simulink.
  - Analyzed drive performance and dynamic response under different operating conditions.
- **Five-Level Inverter with Thermal Modeling** *Wide Bandgap Devices*
  - Modeled and simulated a five-level inverter using PLECS with wide-bandgap power devices.
  - Performed thermal modeling to analyze device temperature and power loss.
- **Grid-Forming Inverter for Microgrids** *Power Converters for Microgrids*
  - Developed a grid-forming inverter model for microgrid applications in MATLAB/Simulink.
  - Simulated inverter control and evaluated voltage and frequency response.
- **Flyback Converter Modeling and Analysis** *Power Electronics*
  - Modeled and simulated a flyback converter in LTspice.
  - Analyzed switching waveforms and output voltage behavior.

CURRENT POSITIONS OF RESPONSIBILITY

- **Teaching Assistant**, M.Tech Power Electronics Laboratory, NIT Warangal
  - Assisted M.Tech students during power electronics laboratory sessions involving converter circuits, switching devices, and experimental procedures.
  - Supported laboratory experiments, circuit troubleshooting, and verification of experimental results.

- **Research Intern**, NIT Warangal
  - Worked on surface-mounted PMSM modeling and analysis using Ansys Maxwell 2D under the guidance of Dr. Sharmili Das.
  - Analyzed electromagnetic machine characteristics through finite element modeling and simulation.

Technical Skills

Developer Tools : dSPACE, MATLAB/Simulink, LTspice, PLECS, KiCad, ANSYS Maxwell.

Embedded Systems : TI C2000 F28379D, ADC, ePWM, interrupts, PI Control, Motor Control Algorithms, CCS.

Programming Languages : C, Embedded C, Python (Basic).

RELEVANT COURSES

Power Electronics	Electric Drives	Control Systems
Advanced Electric Drives	Digital Signal Processor (DSP)	Wide Bandgap Devices
Power Converters for Microgrids	FACTS and HVDC	Electrical Machines

Achievements

•**GATE 2025, 2026** Qualified 2 times - Electrical Engineering *2025–2026*

•**First Prize**Project Expo, SRKR Engineering College *2024*